PRESIDENCY UNIVERSITY

Department of Computer Science and Engineering

CSE324

DATABASE MANAGEMENT SYSTEMS LABORATORY

Course Plan

Course Code	CSE324
Course Type	Laboratory
Credits	01
Co-requisite	CSE323
Number of Classes	10
Laboratory Focus	Relational database design, SQL, and application

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1 Course Overview

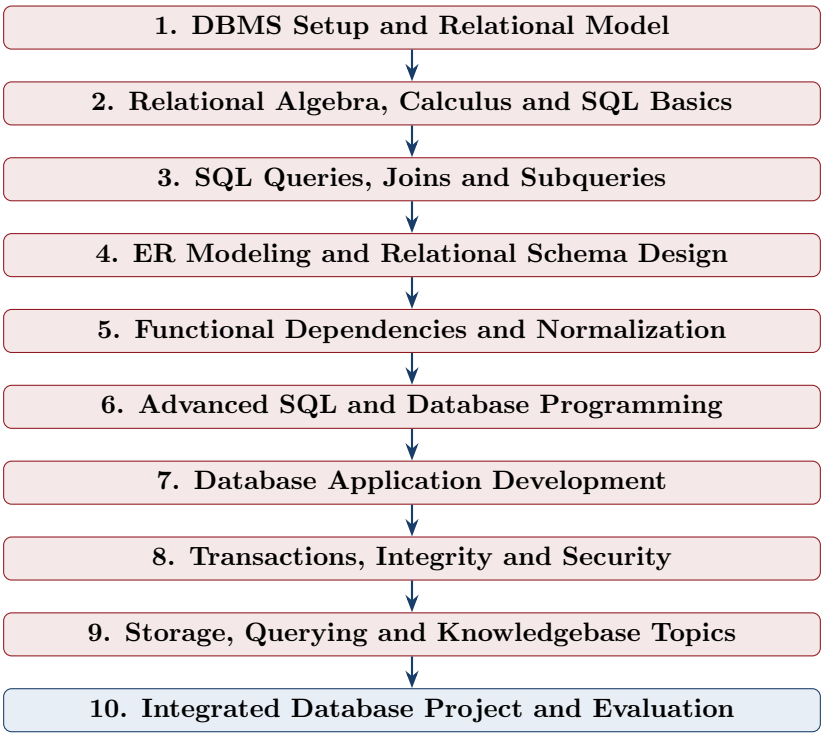
CSE324: Database Management Systems Laboratory provides practical experience in designing, implementing, querying, and maintaining relational databases. The laboratory complements CSE323 by translating database concepts into working schemas, SQL queries, and database-backed applications.

Students progress from database fundamentals and relational operations to conceptual modeling, SQL, schema refinement, transactions, security, storage, and application development. The final activities integrate these skills in a small database project.

1.1 Course Information

Course	CSE324 Database Management Systems Laboratory
Co-requisite	CSE323
Credits	01
Course rationale	Practice in database design and implementation, with emphasis on SQL and front-end application development.

1.2 Course Progression



2 Course Objectives

The course aims to:

1. Apply a systematic methodology to conceptual and relational database design.
2. Implement relational schemas with appropriate keys, constraints, and relationships.
3. Formulate and execute SQL queries for data definition, manipulation, and retrieval.
4. Evaluate schema quality using functional dependencies and normalization.
5. Develop a basic front-end application that interacts with a database.

6. Apply transaction, integrity, and security principles in database applications.
7. Collaborate to design, implement, document, and demonstrate a database project.

3 Course Learning Outcomes (COs)

CO	Learning Outcome
CO1	Explain database design methodology and verify the structural correctness of a database schema.
CO2	Implement relational databases and database application software.
CO3	Use and interpret SQL and supporting database tools to define, manipulate, and retrieve data.
CO4	Explain and apply normalization to improve relational schema quality.
CO5	Collaborate in a group to design, implement, test, and present a database project.

4 Ten-Lecture Course Plan

Class	Topic	Laboratory Contents	Practical Activity / Deliverable	COs
1	DBMS Introduction and Relational Model	Database environment and client tools; database, table, tuple, attribute, domain; relation schemas; primary and foreign keys; entity, referential, and domain constraints.	Install/connect to a DBMS; create a database and related tables with keys and constraints.	CO1, CO2
2	Relational Algebra, Relational Calculus, and SQL Foundations	Selection, projection, union, difference, Cartesian product, joins, and division; declarative query concepts; mapping basic relational operations to SQL.	Express sample requests in relational algebra and implement equivalent SQL queries.	CO1, CO3
3	SQL: Data Definition, Manipulation, and Retrieval	CREATE, ALTER, DROP; INSERT, UPDATE, DELETE; SELECT, WHERE, ORDER BY, DISTINCT; aggregate functions, GROUP BY, HAVING, joins, nested queries, set operations, and views.	Solve a graduated set of queries on a sample database; submit an SQL script.	CO2, CO3
4	ER Modeling and Relational Schema Design	Entities, attributes, relationships, cardinality, participation, weak entities, keys; ER diagrams; mapping ER designs to relational schemas.	Model a real-world scenario (e.g., library or course registration) and implement its relational schema.	CO1, CO2
5	Functional Dependencies and Normalization	Functional dependencies; candidate keys; attribute closure; 1NF, 2NF, 3NF, BCNF; lossless decomposition and dependency preservation.	Analyze a deliberately flawed schema, normalize it, and justify the decomposition.	CO1, CO4
6	Advanced SQL and Database Programming	Constraints, assertions where supported, views, indexes, triggers, stored procedures/functions where supported; parameterized queries and error handling.	Implement selected database-side logic and test constraints, triggers, or procedures.	CO2, CO3, CO4

Class	Topic	Laboratory Contents	Practical Activity / Deliverable	COs
7	Database Application Development	Application architecture; connecting a front end to a database; CRUD operations; form validation; parameterized statements; displaying query results; basic UI design.	Build a small application that supports create, read, update, and delete operations.	CO2, CO3, CO5
8	Transactions, Integrity, and Security	Transaction concepts; ACID properties; COMMIT and ROLLBACK; concurrent updates and isolation overview; constraints; users, roles, privileges, least privilege, SQL injection prevention, backup basics.	Demonstrate a transaction with rollback; configure access privileges and test invalid inputs safely.	CO2, CO3
9	Data Storage, Querying, Knowledge-bases, and Related Topics	Indexing and query plans; storage and access methods; basic query optimization; database–knowledgebase relationship; semantic web and parallel database concepts (overview and demonstration where tools permit).	Compare query behavior with/without an index; inspect an execution plan; short concept exercise on knowledge representation or semantic data.	CO1, CO3
10	Integrated Database Project and Practical Evaluation	Integration of ER design, relational schema, normalization, SQL, integrity, transactions, security, and application interface; testing and documentation.	Group demonstration, project submission, test cases, brief report, and individual viva.	CO1–CO5

Scheduling note: The sequence is designed for ten laboratory meetings. The instructor may adjust the DBMS, programming language, and framework to match available laboratory resources. Stored procedures, triggers, and some constraint features vary by DBMS.

5 Suggested Laboratory Project

Students work in groups to build a small database-backed system, such as a library management system, course registration system, inventory system, or clinic appointment system. Each project should include:

- Requirements and an ER diagram.
- Relational schema with primary keys, foreign keys, and integrity constraints.
- Normalization analysis and rationale for the final schema.
- SQL scripts for schema creation, sample data, and representative queries.
- A simple front-end supporting CRUD operations through parameterized queries.
- Transaction handling, role-based access where supported, and test cases.
- A concise report and group demonstration.

6 Recommended Laboratory Tools

- Relational DBMS: MySQL, PostgreSQL, SQL Server, Oracle, or another department-approved platform.
- SQL client: DBMS command-line client, pgAdmin, MySQL Workbench, DBeaver, or equivalent.

- Application layer: a course-approved language and framework (for example, Python, Java, PHP, or JavaScript).
- ER diagramming tool and version control for group collaboration.

7 Laboratory Report Format

Each submitted experiment or milestone should include:

1. Experiment title, class number, and student/group information.
2. Objective and brief description of the task.
3. Database schema or ER diagram, as applicable.
4. SQL statements and relevant application code.
5. Test data, query output, and screenshots where appropriate.
6. Discussion of results, errors encountered, and corrective steps.
7. Conclusion.

8 Suggested Assessment Scheme

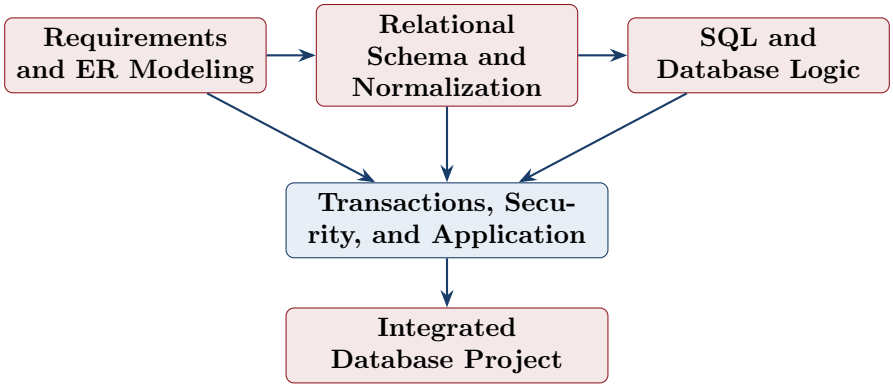
The following is a proposed distribution and may be aligned with the department's approved assessment policy.

Assessment Component	Weight
Attendance and participation	10%
Lab performance / short exercises	10%
SQL and design assignments	20%
Project milestones and report	20%
Final practical examination and viva	40%
Total	**100%**

9 Recommended Books

1. Ramez Elmasri and Shamkant B. Navathe, *Fundamentals of Database Systems*, 6th edition, Pearson, 2010.
2. Raghu Ramakrishnan and Johannes Gehrke, *Database Management Systems*, 3rd edition, McGraw-Hill, 2002.

10 Course Concept Map



Conclusion

CSE324 is organized to help students move from database concepts to practical implementation. Across ten laboratory classes, students design relational schemas, formulate SQL queries, improve schema quality through normalization, and develop a database-backed application. The integrated project reinforces technical skills, testing practices, and collaborative work in alignment with the course learning outcomes.